

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

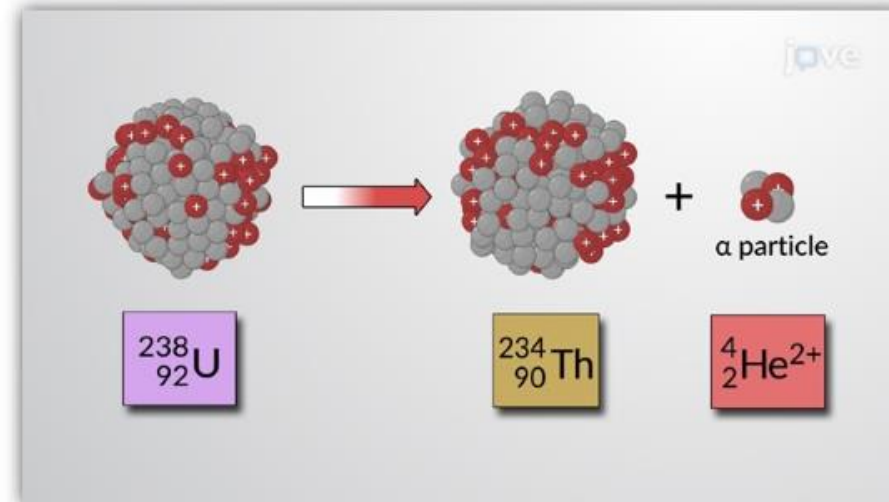
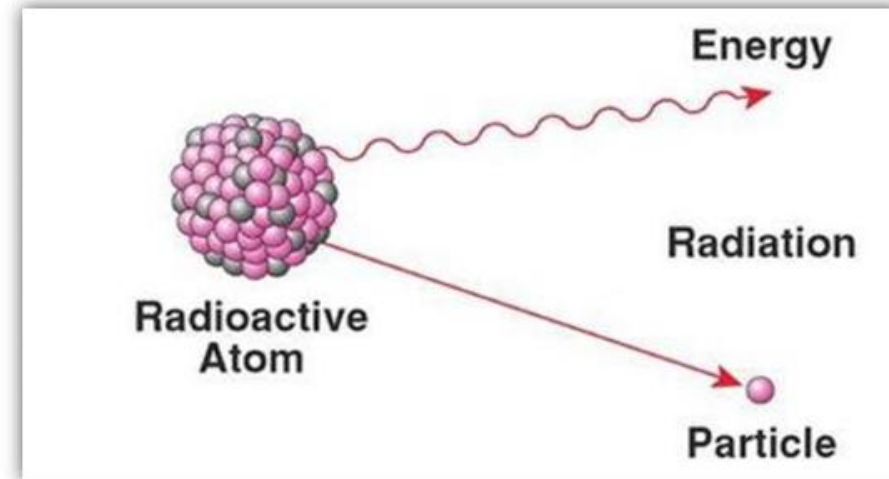
MODULE - 35

NUCLEAR TECHNOLOGY (ICT) - 1

NUCLEAR TECHNOLOGY

RADIOACTIVITY OR RADIOACTIVE DECAY

- Radioactivity is the act of emitting radiation spontaneously. This is done by an atomic nucleus that is unstable. The unstable nucleus wants to give up some energy in order to shift to a more stable state.
- The spontaneous change of an unstable nucleus into another is radioactive decay. The unstable nucleus is called the parent nucleus and the nucleus that results from the decay is known as the daughter nucleus. The daughter nucleus may be stable, or it may decay itself.



NUCLEAR TECHNOLOGY

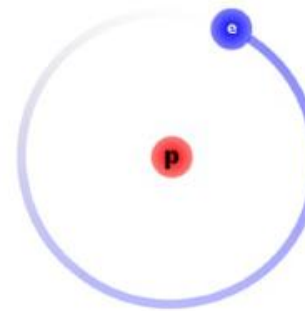
ISOTOPE

DEFINITION 1

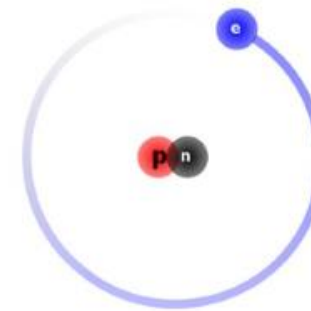
- Isotopes are atoms of the same element that have different numbers of neutrons but the same number of protons and electrons.
- ✓ Due to the difference in the number of neutrons between the various isotopes of an element, they have different masses.

DEFINITION 2

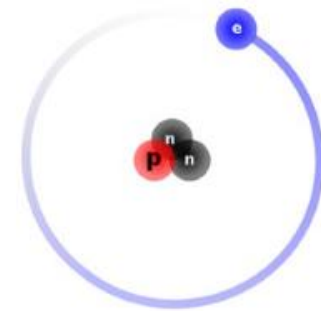
- Isotopes are types of atoms that have the same atomic number (number of protons in their nuclei) and different nucleon numbers (mass numbers) due to different numbers of neutrons in their nuclei.
- ✓ All the isotopes of an element have almost the same chemical properties, but have different atomic masses and physical properties.



Protium



Deuterium

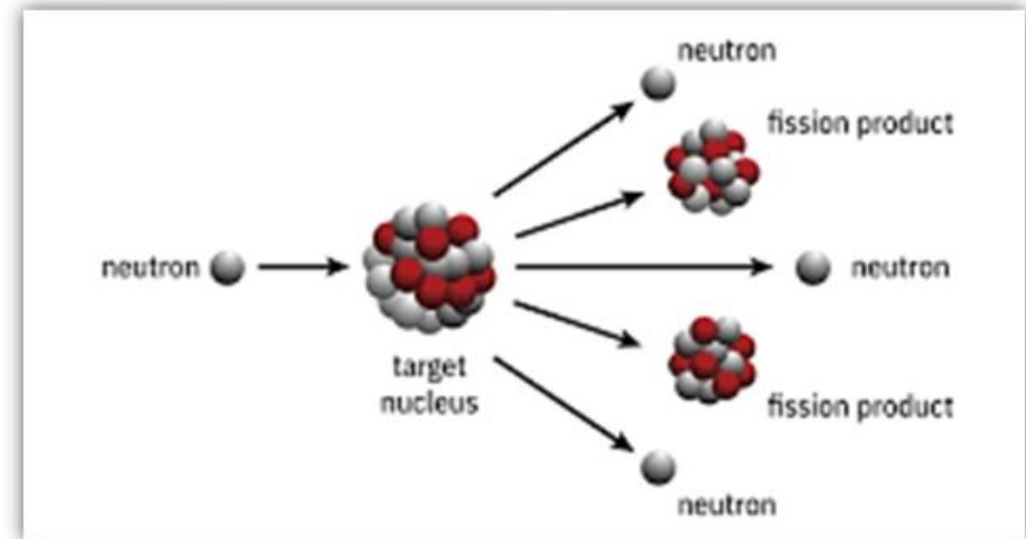


Tritium

Fission and fusion are two processes that produce huge amounts of energy from atoms. They yield millions of times more energy than other sources.

FISSION

- Fission occurs when a neutron collides into a nucleus, forcing it to excite and split into two smaller nuclei (known as fission products). Additional neutrons are also released so that they can initiate a chain reaction.
- When each nucleus splits, a tremendous amount of energy is released.
- Uranium and plutonium are most commonly used for fission reactions in nuclear power reactors because they are easy to initiate and control.
- The energy released by fission in nuclear reactors heats water into steam. The steam is used to spin a turbine to produce carbon-free electricity.

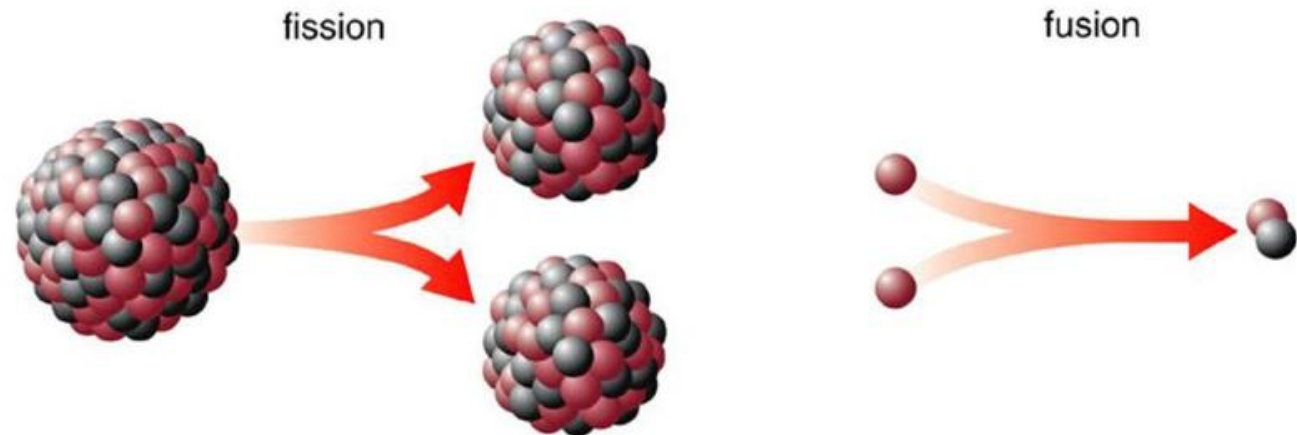


NUCLEAR TECHNOLOGY

FUSION

- Fusion occurs when two nuclei combine together to form a heavier nucleus, Example - two hydrogen nuclei fuse to form one helium nucleus.
- This is the same process that powers the sun and the stars and produces huge amounts of energy, several times greater than fission.
- Fusion reactions are being studied by scientists, but are difficult to sustain for long periods of time because of the tremendous amount of pressure and temperature needed to join the nuclei together.

Fission is the splitting of a heavy, unstable nucleus into two lighter nuclei, and fusion is the process where two light nuclei combine together to form a heavy nucleus and releasing vast amounts of energy.



NUCLEAR TECHNOLOGY

INDIA'S NUCLEAR PROGRAM

- Nuclear energy is most controversial, yet it is critical for India's energy security.
- India's energy demand is expected to rise to 800 GW by 2032, so it is important to consider every source of energy for meeting such a huge demand.
- India has vast thorium reserves; this makes arguments in favour of nuclear energy even more compelling. But there are also strong and reasonable apprehensions against the use of nuclear energy which can't be ignored.
- India's Nuclear Program has roots in the establishment of Tata Institute of Fundamental Research in 1945, under the initiative of Dr Homi Bhabha. TIFR undertook research in nuclear physics.



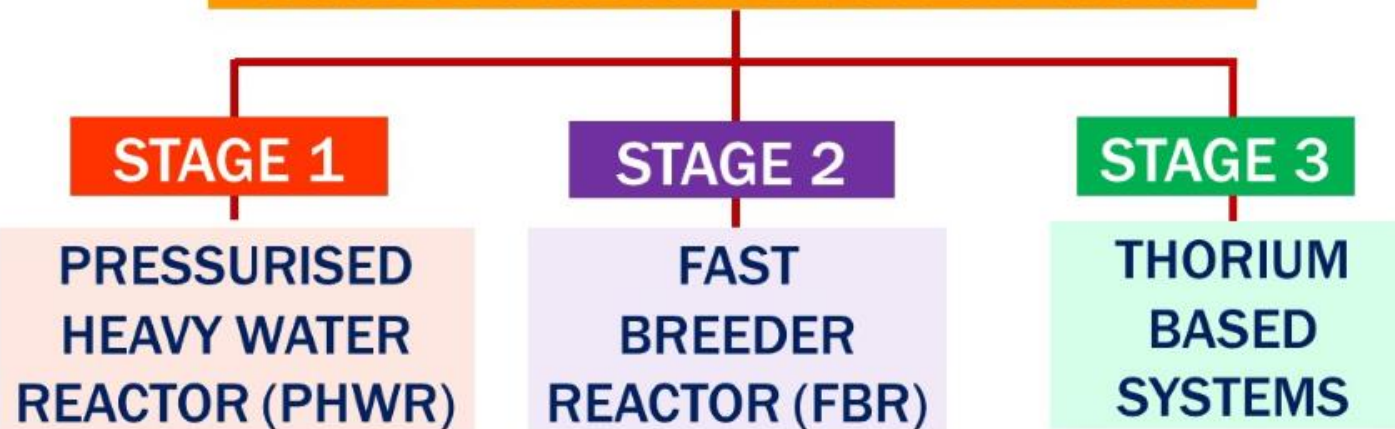
DR. HOMI JEANGIR BHABHA

NUCLEAR TECHNOLOGY

INDIA'S NUCLEAR PROGRAM

- In 1948 the Atomic Energy Commission was established to work for Indian Nuclear Programme. The main work of the Commission was prospecting the nuclear raw materials of the country.
- India has insufficient Uranium reserves (1-2% of global reserves), but is endowed with one of the largest reserves of Thorium (around 30% of global reserves).
- Thorium is not fissile and can't be used directly in nuclear reactors. But it is fertile. Isotope of Thorium, Thorium - 232 can be converted to Uranium - 233 which is fissile. This process of conversion is called Transmutation.
- To exploit India's Thorium reserves, Dr Homi Bhabha conceived a '3 Stage Nuclear Program'.

THREE STAGE NUCLEAR PROGRAM



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